

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Course Code/ Name: DSA0601 /Data Handling and Visualization

Slot C | CO1 – AT3 (R Only)

Q1. Table to Visual: Colour as a tool to highlight [10 marks]

Course	AvgMarks
Chem	
	75
History	
	85
Economics	
	65
Biology	
	60
Physics	
	55
English	
	95
Stats	
	50

A manager wants to draw attention to all courses below the target of 62 among the 7 courses below.

(a) Answer: Highlight Physics (55) and Stats (50), because their average marks are below 62. Highlighting makes the important exceptions immediately visible.

(b) Answer: Use crimson/red (#DC143C) for highlighted bars and light grey (#D3D3D3) for other bars, with a dashed line at 62.

```
library(ggplot2)
```

```
df <-
```

```
data.frame(Course=c("Chem","History","Economics","Biology","Physics","English","Stats"),
```

```
          AvgMarks=c(75,85,65,60,55,95,50))
```

```
df$Highlight <- df$AvgMarks < 62
```

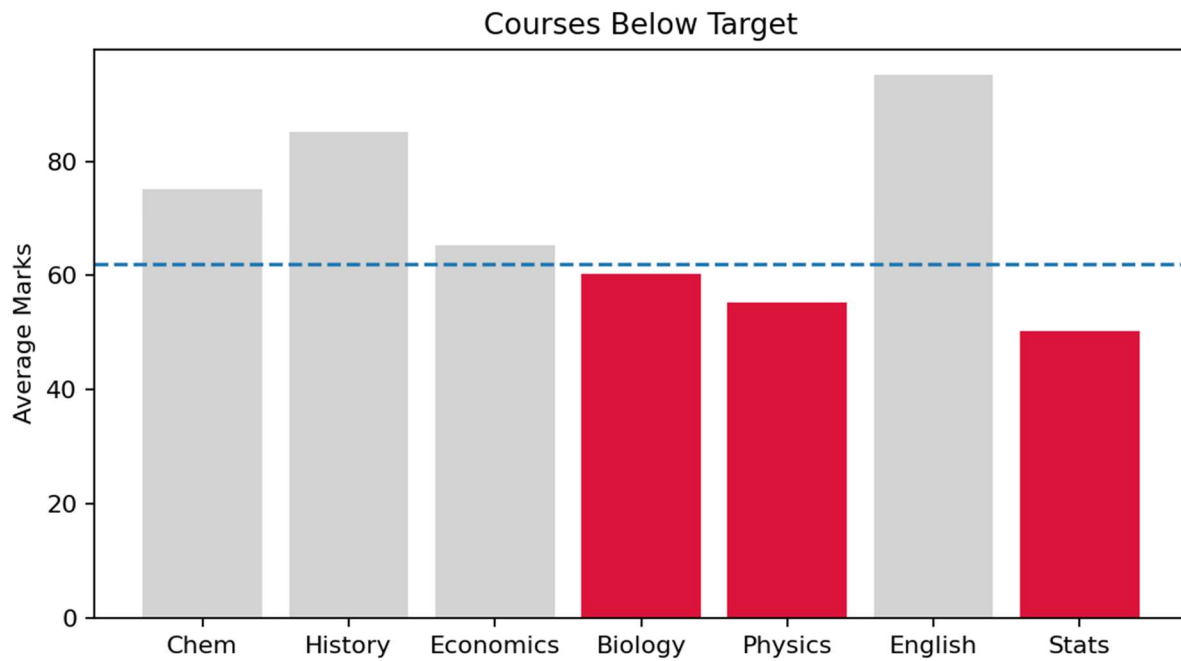
```
ggplot(df,aes(Course,AvgMarks,fill=Highlight)) +
```

```
  geom_col() +
```

```
  scale_fill_manual(values=c("FALSE"="#D3D3D3","TRUE"="#DC143C")) +
```

```
  geom_hline(yintercept=62,linetype="dashed") +
```

```
  theme_minimal() + theme(legend.position="none")
```



(c) Answer: Accent/highlight scheme. A rainbow palette is poor because arbitrary hue changes can imply false categories and are less accessible.

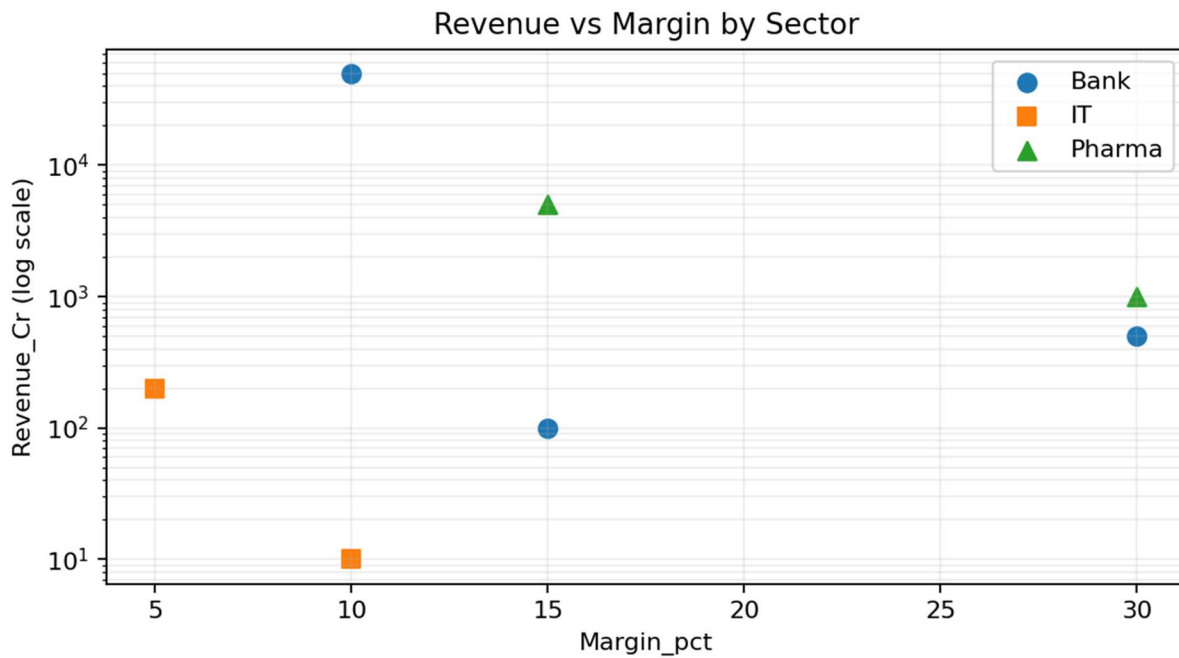
(d) Answer: Use a colour-blind-safe palette and add a non-colour cue such as labels, patterns, or a bold outline.

Q2. Visual to R Code: Aesthetics and a log axis [10 marks]

Margin_pct	Revenue_Cr	Sector
10		
	10	
		IT
30		
	500	
		Bank
15		
	100	
		Bank
30		
	1000	
		Pharma
15		
	5000	
		Pharma
10		
	50000	
		Bank
5		
	200	
		IT

(a) Complete R program:

```
library(ggplot2)
df <- data.frame(
  Margin_pct=c(10,30,15,30,15,10,5),
  Revenue_Cr=c(10,500,100,1000,5000,50000,200),
  Sector=c("IT","Bank","Bank","Pharma","Pharma","Bank","IT"))
ggplot(df,aes(x=Margin_pct,y=Revenue_Cr,colour=Sector)) +
  geom_point(size=3) +
  scale_y_log10() +
  labs(x="Margin_pct",y="Revenue_Cr (log scale)")
```



(b) Answer: Margin_pct → x position; Revenue_Cr → y position; Sector → colour.
Revenue_Cr needs a log axis because it ranges from 10 to 50,000, spanning several orders of magnitude; a log scale prevents small values from being compressed.

Q3. Design Decision: Choosing aesthetics and a colour palette [10 marks]

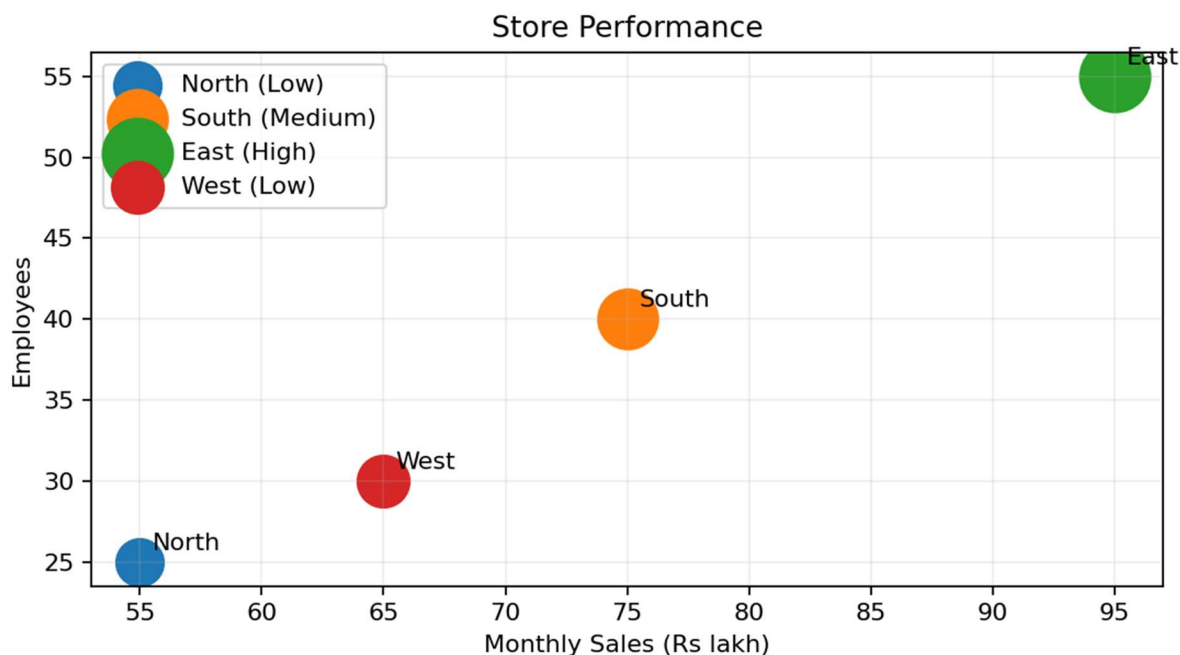
Variable	Description
Region	
	North / South / East / West
Monthly_Sales	
	monthly sales in Rs lakh
Employees	
	number of employees
Customer_Satisfaction	
	Low / Medium / High

(a) Answer: Region = nominal; Monthly_Sales = continuous; Employees = discrete quantitative; Customer_Satisfaction = ordinal.

(b) Answer: Monthly_Sales → x position (continuous); Employees → y position (continuous numerical display of counts); Customer_Satisfaction → colour (ordered/sequential); Region → shape (discrete/qualitative).

(c) Answer: Use a sequential palette for Customer_Satisfaction because Low < Medium < High. In R, scale_colour_viridis_d() is a suitable colour-blind-friendly ordered palette.

```
ggplot(store_data,
  aes(x=Monthly_Sales,y=Employees,
    colour=Customer_Satisfaction,shape=Region)) +
  geom_point(size=4) +
  scale_colour_viridis_d() + theme_minimal()
```



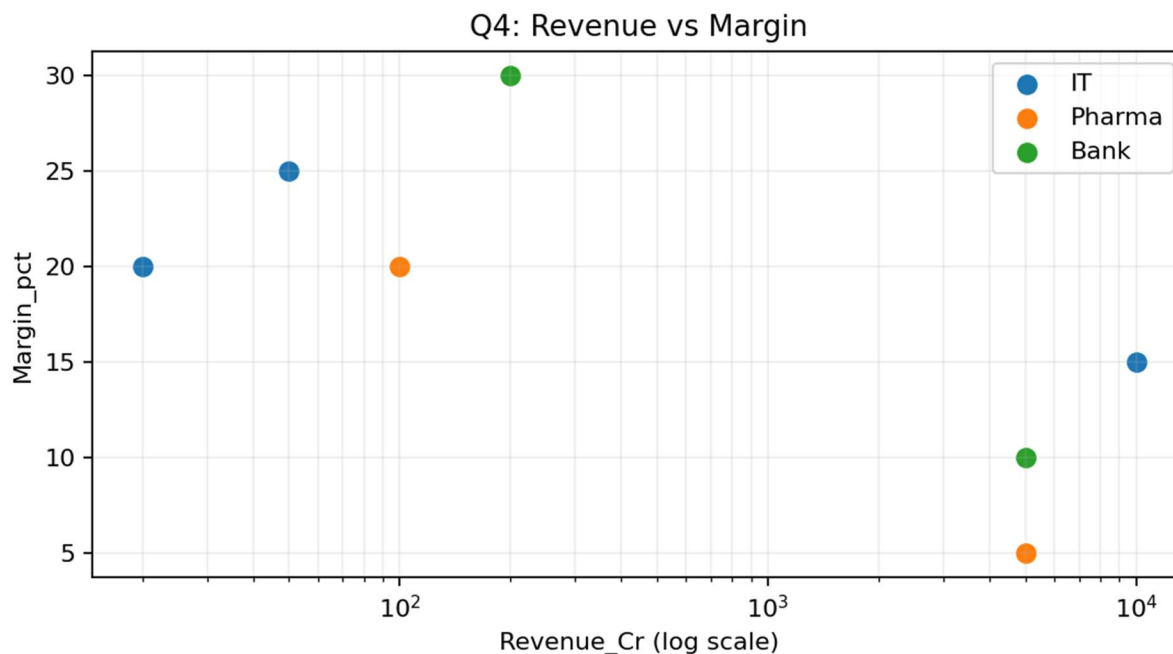
(d) Answer: Use a colour-blind-safe palette such as Viridis, and add redundant encodings such as shape, labels, or direct annotations so colour is not the only cue.

Q4. R Code to Visual: Aesthetics and a log axis [10 marks]

Given R program:

```
library(ggplot2)
df <- data.frame(
  Name=c("Gamma","Zeta","Sigma","Theta","Alpha","Omega","Kappa"),
  Revenue_Cr=c(20,5000,10000,5000,50,100,200),
  Margin_pct=c(20,5,15,10,25,20,30),
  Sector=c("IT","Pharma","IT","Bank","IT","Pharma","Bank"))
ggplot(df,aes(x=Revenue_Cr,y=Margin_pct,colour=Sector)) +
  geom_point(size=3) + scale_x_log10() +
  labs(x="Revenue_Cr (log scale)")
```

(a) Answer: The x-axis is logarithmic Revenue_Cr; the y-axis is linear Margin_pct. Points are Gamma (20,20,IT), Zeta (5000,5,Pharma), Sigma (10000,15,IT), Theta (5000,10,Bank), Alpha (50,25,IT), Omega (100,20,Pharma), Kappa (200,30,Bank).



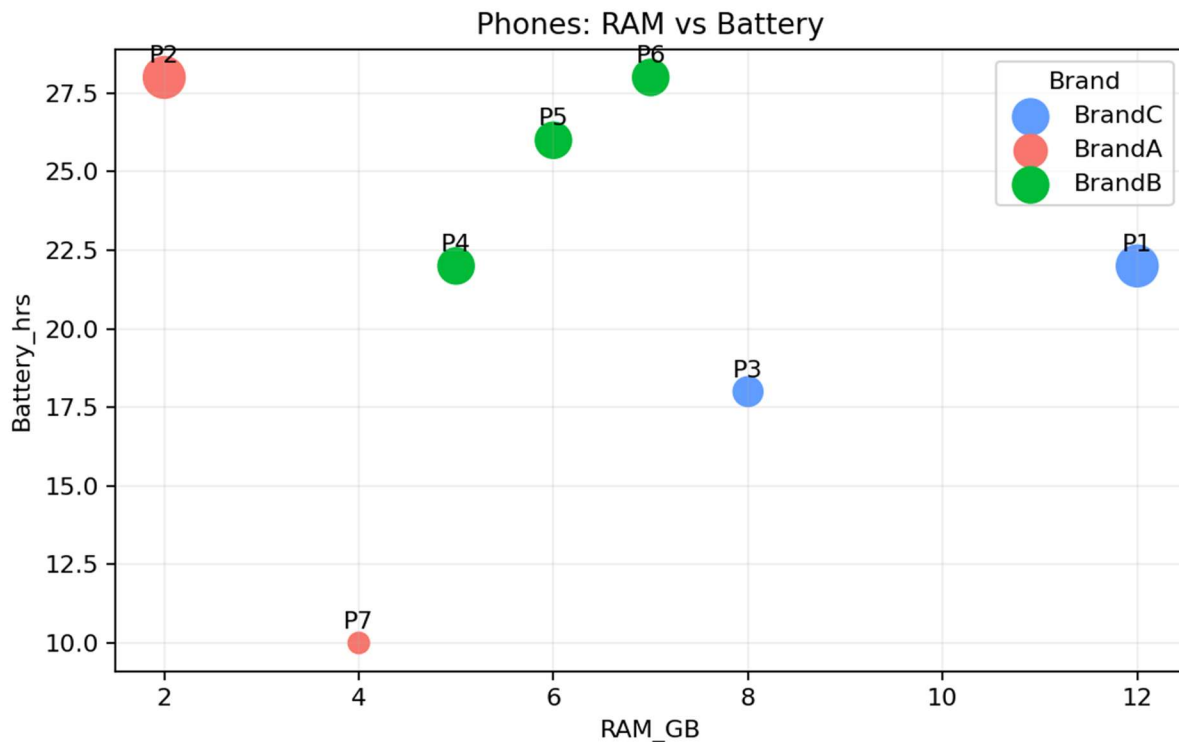
(b) Answer: Revenue_Cr → x position, continuous log scale; Margin_pct → y position, continuous linear scale; Sector → colour, discrete qualitative scale; coordinate system = Cartesian.

(c) Answer: Replace the default colour scale with:
`scale_colour_viridis_d()` # colour-blind-safe Viridis palette

Q5. Visual to Table: Reading aesthetics back into data [10 marks]

(a) Reconstructed table:

ID	RAM_GB	Battery_hrs	Brand	Price_K
P1				
	12			
		22		
			BrandC	
				40
P2				
	2			
		28		
			BrandA	
				40
P3				
	8			
		18		
			BrandC	
				20
P4				
	5			
		22		
			BrandB	
				30
P5				
	6			
		26		
			BrandB	
				30
P6				
	7			
		28		
			BrandB	
				30
P7				
	4			
		10		
			BrandA	
				10



(b) Answer: RAM_GB $\rightarrow$ x position, continuous; Battery_hrs $\rightarrow$ y position, continuous; Brand $\rightarrow$ colour, discrete/qualitative; Price_K $\rightarrow$ point size, continuous. ID is an identifier/label.

(c) Answer: ID = nominal identifier; RAM_GB = continuous numerical; Battery_hrs = continuous numerical; Brand = nominal categorical; Price_K = continuous numerical.

Final Answer Summary

Question	Key answer
Q1	
	Physics & Stats highlighted below 62
Q2	
	Margin x, Revenue log-y, Sector colour
Q3	
	Satisfaction sequential colour, Region shape
Q4	
	Revenue log-x, Margin linear-y, Sector colour
Q5	
	RAM x, Battery y, Brand colour, Price size